Wind Generator Design

Project Overview

The goal is to generate a controlled wind field over a stratified two-layer water column in the existing 33 m glass wave channel. Target conditions:

- Wind speed: 10 m/s (steady-state target)
- Wind acceleration: $0.16\,\mathrm{m/s^2}$ (ramp from 0 to 10 m/s over 60 s)
- Water column: saltwater ≈44 cm + freshwater ≈10 cm on top (stable density stratification)

Repeatability is a primary design driver. Each experimental run requires **straight, uniform flow** with **known turbulence intensity**, a **precisely timed and reproducible wind start**, and a **slow, controlled acceleration** so that all instruments can be triggered simultaneously from a common start signal.

Previous Experimental Cases

		Freshwater	**Saltwater + 5 cm fresh layer**	**Saltwater + 10 cm fresh layer**
h_{air}		0.35 m	0.35 m	0.33 m
h_{water}	h_{fresh}	0.49 m	0.05 m	0.10 m
	h_{salt}	0.00 m	0.44 m	0.44 m

Instrumentation Suite

Multiple diagnostics operate simultaneously following wind onset:

- Surface temperature: infrared (IR) imaging
- Subsurface mixing: laser-induced fluorescence (LIF)
- Subsurface velocity: particle image velocimetry (PIV)
- Air-side turbulence statistics: hot-wire anemometry
- Surface wave slope field: color imaging slope gauge (CISG)

Configuration

Suction-mode **open-channel** wind tunnel on the existing 33 m glass wave tank. The fan at the beach end pulls ambient laboratory air through the inlet assembly at the upstream (wave-maker) end. There is no recirculating return path. A removable lid covers the full length of the channel to form the test-section ceiling. A wave-maker cage at the upstream end of the tank is the primary geometric obstacle that the S-curve must route around.

High-level flow path:

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Bell-mouth -> Settling chamber (honeycomb + screens) -> Contraction 4:1
  -> S-curve down into tank -> Adjustable plate at S-curve exit
  -> 33 m fetch over water -> Beach / diffuser -> Fan
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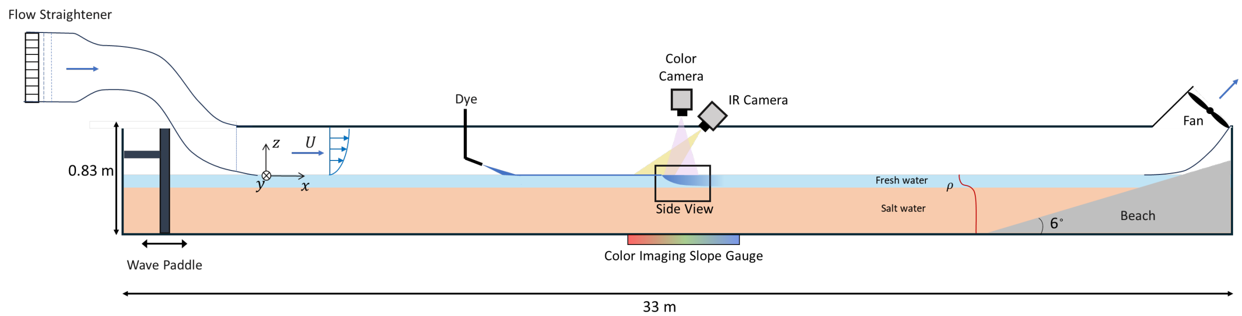

Figure 1:

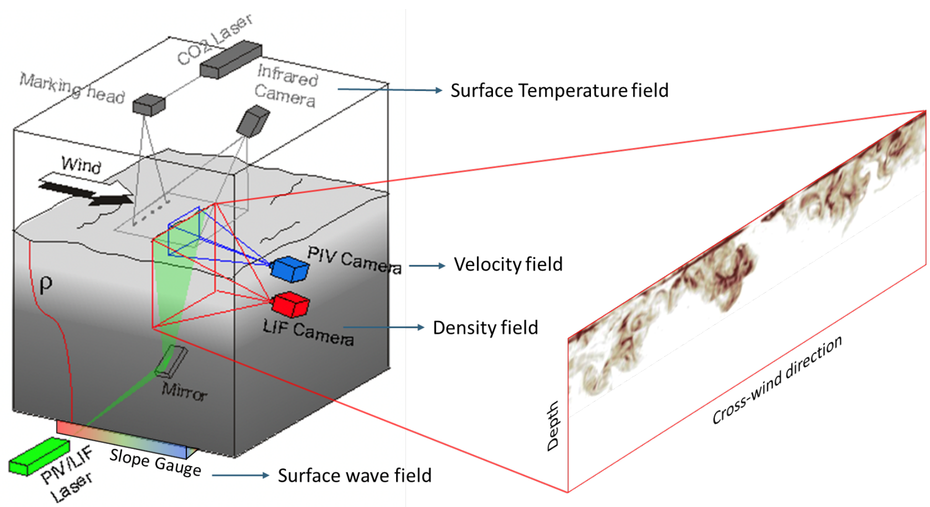

Figure 2:

Figure 3:

Current Design Dimensions

Item	Dimension
Water level	490-540 mm
Tank bottom to lid ceiling	840 mm
Test section (W × H)	500 × 330 mm, 33 m long
S-curve cross-section	500 × 330 mm constant
S-curve length	1589 mm (Set by R_{min} criterion)
S-curve vertical drop	550 mm from contraction exit to adjustable plate (±10 mm above MWL)
S-curve min radius	796 mm ($\approx 2.0\, D_h$)
Contraction ratio	4:1, max wall angle 7°
Contraction entry (W × H)	902 × 732 mm
Contraction exit (W × H)	500 × 330 mm
Contraction length	≥2200 mm
Contraction wall profile	5th-order polynomial
Settling Chamber same as contraction entry	902 × 732 mm
Settling Chamber length	≥1200 mm (enough for honeycomb and 2-3 screens)
Settling Chamber internal velocity	≤3 m/s (at least 25% of test section velocity)
Bell Mouth same as contraction entry	902 × 732 mm
Bell Mouth entry lip elliptical profile	$a = 120$ mm, $b = 30$ mm ($\zeta \leq 0.05$)
Diffuser axis tilt	6° (aligned with beach slope)
Diffuser entry	500 × 330 mm
Diffuser exit	686 × 686 mm
Diffuser length	≈1000 mm along beach axis
Diffuser screen	40–50% open area at $t \approx 0.4$ (required)
In-line duct fan	inlet diameter 24 in, outer duct box 27 × 27 × 25.5
Target wind speed	10 m/s but higher the better in case we lower the water level

Sheet Metal

- Body: 20-gauge (0.9 mm) galvanized steel
- Hat-channel stiffeners on flat panels >300 mm
- 3 mm flat bar flanges at all bolted joints
- Lower S-curve piece fits inside the glass channel: $500 - 2 \times 0.9 = 498.2$ mm clear width
- Screen frame: removable bolted tabs for salt-clog maintenance

Fan

- Dayton 852H87, 2 HP, mixed flow, 20 in wheel

- Voltage: 208–240 V single phase (run the cable to 240V outlet)
- ECM motor with built-in variable speed controller
- Speed control via 0–10 V analog input from DAQ (LabVIEW) —
- Fan housing (27 in × 27 in × 25.5 in box) mounts flush on diffuser exit, inclined at beach angle

Flow Conditioning (Honeycomb + Screens)

Screen	Position from inlet	Wire dia. d_w	Open area β	K approx.
Honeycomb (already have)	0-50 mm	5–8 mm hex cells		
1 (coarse)	250 mm	0.8 mm	0.40	∼1.0
2 (medium)	700 mm	0.5 mm	0.38	∼1.0
3 (fine)	contraction exit face	0.3 mm	0.35	∼0.8

Minimum spacing between screens: $\geq 500 \times d_w$ (i.e. ≥400 mm after Screen 1, ≥250 mm after Screen 2).

Screen loss coefficient estimated as:

$$K_{\text{screen}} = \frac{(1-\beta)^2}{\beta^2 C_d^2}, \qquad C_d \approx 0.6 \tag{1}$$

Screen 3 can be place at the contraction exit or after the S-curve

Contraction

CR = 4:1 was chosen based on standard wind tunnel design practice. The Delaware and Miami facilities were used as configuration benchmarks, but neither publishes detailed design specifications, so their exact contraction ratios are unknown. The minimum CR for acceptable flow uniformity is ≈3:1; 4:1 provides a comfortable margin without making the settling chamber impractically large. A higher CR (e.g. 6:1) would improve turbulence suppression but was ruled out due to available space, fabrication cost, and the requirement that the assembly be easily removable when the wind tunnel is not in use. CR = 5:1 would require an 827 mm tall settling chamber entry.

The 7° wall angle is a hard separation limit. Entry dimensions are set by the **equal wall angle method**: height and width are each expanded by the same absolute amount Δ so both wall angles are equal at the maximum permissible value.

$$\Delta = \frac{-830 + \sqrt{830^2 + 4 \times 165{,}000 \times (CR-1)}}{2} = 402\,\text{mm} \tag{2}$$

Wall Profile (5th-order polynomial)

$$y(t) = y_1 - (y_1 - y_2)\left[6t^5 - 15t^4 + 10t^3\right], \quad t \in [0,1] \tag{3}$$

S-Curve Transition

The S-curve routes the flow around the wave-maker cage and drops the duct exit to near mean water level (MWL). It is a constant cross-section (500 × 330 mm) turning duct; its job is positioning, not acceleration.

Note: the 7° wall angle limit does not apply for the S-curve, because it's a constant-area turning duct, not a diffuser. The governing criterion is $r/D_h \geq 1.5$–2.0 (curvature-based, not angle-based).

Adjustable Plate at S-Curve Exit

Flat plate, 500 mm wide, slot-and-bolt mount, ±20 mm vertical adjustment, 10–15° ramp from 10 mm down to the mean water level

Diffuser / Beach Transition

This section is not finished. I didn't know how to handle the asymmetric wall expansion with the beach angle. Below are the requirements and what's been established so far.

The diffuser recovers dynamic pressure from the test section back into static pressure, reducing the load on the fan. Without it, the fan works against the full dynamic pressure of a 10 m/s stream.

Using the beach slope as the diffuser floor is convenient but produces **fully one-sided area growth**: only the ceiling and side walls expand while the floor (beach) is fixed. Fully asymmetric diffusers are significantly more prone to flow separation than symmetric ones. A flow-conditioning screen at ≈40–50% open area placed at $t \approx 0.4$ along the diffuser length is therefore **non-negotiable** to redistribute momentum and suppress separation.

Options Under Consideration

1. **Accept asymmetric expansion + screen.** Simplest fabrication. Beach is the floor. Ceiling and side walls expand via 5th-order polynomial lofts. Screen at mid-diffuser.
2. **Short S-curve + symmetric diffuser.** Add a small S-curve at the test-section exit to bring the duct centerline off the beach surface, then run a more symmetric diffuser above the beach. Longer overall length.
3. **Raise the diffuser floor early.** Start the floor angle steeper than the beach to gain area on both sides faster, then relax. Keeps the 7° per-wall limit.

Parameter	Value	
Entry	500 × 330 mm	
Exit	686 × 686 mm (Matches fan housing face)	
Area ratio	∼2.85:1	—
Beach slope	$\approx 6 - 9°$	
Diffuser axis	Aligned with beach slope (Beach act as the diffuser floor)	
Ceiling expansion	330 → 686 mm (5th-order polynomial profile or linear?)	
Screen in the middle of diffuser?	40–50% open area	

Wind Speed Estimate

This is a rough estimate.

Operating point found by matching system pressure loss to fan curve:

$$\Delta P_{\text{system}}(U) = \Delta P_{\text{fan}}(Q), \qquad Q = A \cdot U = 0.165\,U \tag{4}$$

System losses summed as:

$$\Delta P_{\text{system}} = \sum_i K_i \cdot \tfrac{1}{2}\rho U_i^2 \tag{5}$$

where each K_i is referenced to the local dynamic pressure $\frac{1}{2}\rho U_i^2$.

Loss Budget

Component	K	Referenced to	ΔP at $U = 10\,\text{m/s}$ (Pa)
Bell-mouth	0.04	q_{inlet}	∼2
Honeycomb	0.60	q_{chamber}	∼14
Screen 1 ($\beta = 0.40$)	1.00	q_{chamber}	∼24
Screen 2 ($\beta = 0.38$)	1.00	q_{chamber}	∼24
Screen 3 ($\beta = 0.35$)	0.80	q_{test}	∼29
Contraction	0.05	q_{test}	∼3
S-curve bend ($r/D_h \approx 2$)	0.15	q_{test}	∼9
Test section friction	$f(L/D_h)$	q_{test}	∼25
Diffuser + screen	0.40	q_{test}	∼24
Total	$K_{\text{total}} \approx 4.5$	—	**∼ 154 Pa**

At $U = 10\,\text{m/s}$: $\Delta P \approx 154\,\text{Pa} \approx 0.62\,\text{in WC}$.

Fan curve for Dayton 852H87: delivers ≈5,630 CFM at 0.50 in WC and ≈5,853 CFM at 0.25 in WC.

Required flow at 10 m/s: $Q = 0.165 \times 10 = 1.65\,\text{m}^3/\text{s} = 3{,}500\,\text{CFM}$.

Result: The 852H87 operating point at 3,500 CFM sits well within its stable operating range at ≈0.62 in WC system resistance. This gives comfortable headroom — the fan can drive significantly more flow, suggesting wind speeds of **12–15 m/s** are achievable depending on actual system losses.

$$\boxed{U_{\text{est}} \approx 12\text{–}15\,\text{m/s} \quad (\pm 20\text{–}30\%)} \tag{6}$$

Design References

- Mehta, R.D. & Bradshaw, P. (1979). Design rules for small low-speed wind tunnels. *Aeronautical Journal*, 83, 443–449. [Flow conditioning: honeycomb and screen specifications]
- Morel, T. (1975). Comprehensive design of axisymmetric wind tunnel contractions. *Journal of Fluids Engineering*, 97(2), 225–233. [Contraction profile design]
- Idelchik, I.E. *Handbook of Hydraulic Resistance*, 3rd ed. [Pressure loss coefficients K used throughout the loss budget]

- Delaware Air–Sea Interaction Laboratory (ASILF), University of Delaware. Primary facility reference ($42\,\mathrm{m} \times 1\,\mathrm{m} \times 1.25\,\mathrm{m}$ tank, $\mathrm{CR} = 4{:}1$ recirculating wind tunnel). No detailed design paper available; used as configuration benchmark.